iii. Fluorine can't act as a central metal atom being the smallest one

iv. Due to high electronegativity with small size fluorine helps the central atom to attain high coordination number

v. They can undergo the auto ionization.

$$2\ ICl \rightleftharpoons I^{+} + ICl_2^{-}$$

$$2\ ICl_3 \rightleftharpoons ICl_2^{+} + ICl_4^{-}$$

vi. They are strong oxidizing agents

Reaction with alkali:

When heated with the alkalis, larger halogen form oxyhalogens and the smaller forms halide.

$$BrF_5 \xrightarrow{^{-}OH} 5F^{-} + \underset{\text{Bromate ion}}{BrO_3^{-}}$$

$$ICl \xrightarrow{^{-}OH} Cl^{-} + \underset{\text{Hypo iodite ion}}{OI^{-}}$$

Structure of inter halogen compounds:

The structures of different type of interhalogen compunds can be easily explained using VSEPR theory. The details are given below.

Table 3.7

Type	Structure	Hybridisation	bond pairs / lone pairs
AX	Linear	sp^3	1 / 3
AX_3	T shaped	sp^3d	3 / 2
AX_5	Square pyrimidal	sp^3d^2	5 / 1
AX_7	Pentagonal bipyramidal	sp^3d^3	7 / 0

3.3.5 Oxides of halogen

Fluorine reacts readily with oxygen and forms difluorine oxide (F_2O) and difluorine dioxide (F_2O_2) where it has a -1 oxidation state. Other halogens do not react with oxygen readily. But the following oxides can be prepared by some indirect methods. Except fluorine all the other halogens have positive oxidation states.

Table 3.8

Type	X_2O	XO_2	XO_3	X_2O_6	X_2O_7	Others
Oxidation state	+1	+4	+5	+6	+7	-
F	-	-	-	-	-	OF_2 (-1) O_2F_2 (-1) O_4F_2 (-1)
Cl	Cl_2O	ClO_2	-	Cl_2O_6	Cl_2O_7	Cl_2O_4 (+4)

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